so the question of what really happens at their central point is one of science, not speculation. Once again, the equations of general relativity break down under such extreme conditions. Some physicists have suggested that there really is a puncture, but that we are protected from this cosmic "singularity" by the event horizon of the black hole, which prevents anything from escaping its gravitational grip. This reasoning led Roger Penrose of Oxford University to speculate on a "cosmic censorship hypothesis" that allows these kinds of spatial irregularities to occur only if they are deeply hidden from our view behind the shroud of an event horizon. On the other hand, prior to the discovery of string theory, some physicists surmised that a proper merger of quantum mechanics and general relativity would show that the apparent puncture of space is actually smoothed out—"sewn up," so to speak—by quantum considerations.

With the discovery of string theory and the harmonious merger of quantum mechanics and gravity, we are finally poised to study these issues. As yet, string theorists have not been able to answer them fully, but during the last few years closely related issues have been solved. In this chapter we discuss how string theory, for the first time, definitively shows that there are physical circumstances—differing from wormholes and black holes in certain ways—in which the fabric of space can tear.

A Tantalizing Possibility

In 1987, Shing-Tung Yau and his student Gang Tian, now at the Massachusetts Institute of Technology, made an interesting mathematical observation. They found, using a well-known mathematical procedure, that certain Calabi-Yau shapes could be transformed into others by puncturing their surface and then sewing up the resulting hole according to a precise mathematical pattern.2 Roughly speaking, they identified a particular kind of two-dimensional sphere—like the surface of a beach ball—sitting inside an initial Calabi-Yau space, as in Figure 11.2. (A beach ball, like all familiar objects, is three-dimensional. Here, however, we are referring solely to its surface; we are ignoring the thickness of the material from which it is made as well as the interior space it encloses. Points on the beach ball's surface can be located by giving two numbers—"latitude" and "longitude"—much as we locate points on the earth's surface. This is